

COGNIZANT DIGITAL NURTURE 5.0 — JAVA FSE

Deep-Skilling MCQ — Practice Question Bank

Built from the disclosed masterclass topic list (Design Patterns · Spring/Maven/Hibernate · Spring Boot & REST · React · Git · Testing · Core Java · SQL). This is an original practice set.

Core Java & OOP

Q1. Which keyword prevents a method from being overridden in a subclass?

- A. static
- B. final
- C. abstract
- D. private

Q2. What is the output of $5 / 2$ in Java when both operands are int?

- A. 2.5
- B. 2
- C. 3
- D. Compile error

Q3. Which collection class does NOT allow duplicate elements?

- A. ArrayList
- B. LinkedList
- C. HashSet
- D. Vector

Q4. Which of these is NOT a valid access modifier in Java?

- A. public
- B. protected
- C. friend
- D. private

Q5. What does the 'static' keyword mean when applied to a variable?

- A. It belongs to each instance separately
- B. It belongs to the class, shared across all instances
- C. It cannot be changed
- D. It is thread-local

Q6. Which interface must a class implement to be used in a for-each loop?

- A. Comparable
- B. Iterable
- C. Serializable
- D. Cloneable

Q7. What causes a deadlock in Java threads?

- A. Too many objects on the heap
- B. Two or more threads waiting on locks the other holds
- C. A single thread running too long
- D. Garbage collection pausing execution

Design Patterns

- Q8.** The Singleton pattern ensures:
- A. Multiple instances can coexist
 - B. Only one instance of a class exists with global access
 - C. Objects are created via a factory
 - D. Classes are decoupled from implementations
- Q9.** Which pattern is used when you need to create objects without specifying their exact class?
- A. Observer
 - B. Factory
 - C. Decorator
 - D. Adapter
- Q10.** The Observer pattern is best suited for:
- A. One-time object creation
 - B. Notifying multiple dependents when an object's state changes
 - C. Restricting class instantiation
 - D. Wrapping incompatible interfaces
- Q11.** What problem does the Adapter pattern solve?
- A. Reduces memory usage
 - B. Allows incompatible interfaces to work together
 - C. Ensures single instantiation
 - D. Improves loop performance
- Q12.** Which pattern adds new behavior to an object dynamically without altering its class?
- A. Decorator
 - B. Singleton
 - C. Builder
 - D. Proxy
- Q13.** The Builder pattern is most useful when:
- A. An object has many optional constructor parameters
 - B. You need only one instance
 - C. Two classes are incompatible
 - D. You want lazy initialization

Spring Core, Maven & Hibernate

- Q14.** What is Dependency Injection in Spring?
- A. A way to inject SQL queries
 - B. A design pattern where objects receive dependencies rather than creating them
 - C. A Maven plugin
 - D. A Hibernate caching strategy
- Q15.** In Maven, which file defines project dependencies and build configuration?
- A. web.xml
 - B. pom.xml
 - C. build.gradle
 - D. application.properties

- Q16.** What does the Hibernate 'session' represent?
- A. A database connection pool
 - B. A single unit of work between the application and the database
 - C. A REST endpoint
 - D. A Spring bean scope
- Q17.** Which Spring annotation marks a class as a Spring-managed bean generically?
- A. @Controller
 - B. @Repository
 - C. @Component
 - D. @Service
- Q18.** What is Hibernate's first-level cache scoped to?
- A. The entire application
 - B. The Session
 - C. The SessionFactory
 - D. The database
- Q19.** What Maven command compiles and packages a project into a JAR/WAR?
- A. mvn clean
 - B. mvn install
 - C. mvn package
 - D. mvn validate

Spring Boot & Spring REST

- Q20.** What is the main advantage of Spring Boot over plain Spring?
- A. It removes the need for Java
 - B. Auto-configuration and embedded servers reduce boilerplate setup
 - C. It replaces Hibernate entirely
 - D. It only works with MongoDB
- Q21.** Which annotation combines @Controller and @ResponseBody?
- A. @RestController
 - B. @Service
 - C. @Repository
 - D. @Configuration
- Q22.** Which annotation is used to inject values from application.properties?
- A. @Autowired
 - B. @Value
 - C. @Bean
 - D. @Qualifier
- Q23.** What HTTP method does @PostMapping correspond to?
- A. GET
 - B. POST
 - C. PUT
 - D. DELETE

Q24. What does @Autowired do?

- A. Marks a class as an entity
- B. Automatically injects a matching Spring bean
- C. Starts the embedded server
- D. Configures the database URL

Q25. Which starter dependency would you add for building REST APIs in Spring Boot?

- A. spring-boot-starter-data-jpa
- B. spring-boot-starter-web
- C. spring-boot-starter-test
- D. spring-boot-starter-security

React

Q26. What is JSX?

- A. A CSS preprocessor
- B. A syntax extension letting you write HTML-like code in JavaScript
- C. A state management library
- D. A testing framework

Q27. Which hook is used to manage state in a functional component?

- A. useEffect
- B. useState
- C. useContext
- D. useRef

Q28. What triggers a React component to re-render?

- A. Changing a local variable
- B. A change in state or props
- C. Refreshing the browser only
- D. Calling console.log

Q29. What is the purpose of useEffect?

- A. To define component styling
- B. To perform side effects like data fetching after render
- C. To create new components
- D. To bind event listeners only on click

Q30. In React, what are 'props'?

- A. Internal mutable state
- B. Read-only data passed from parent to child components
- C. CSS class names
- D. Lifecycle methods

Q31. What does the virtual DOM primarily improve?

- A. Security
- B. Rendering performance by minimizing direct DOM updates
- C. Network latency
- D. Database queries

Git

Q32. Which command stages changes for the next commit?

- A. git commit
- B. git add
- C. git push
- D. git merge

Q33. What does 'git clone' do?

- A. Creates a new branch
- B. Copies a remote repository to your local machine
- C. Deletes a repository
- D. Merges two branches

Q34. Which command shows the commit history?

- A. git status
- B. git log
- C. git diff
- D. git branch

Q35. What is the difference between 'git merge' and 'git rebase'?

- A. They are identical
- B. Merge preserves history with a merge commit; rebase rewrites commit history onto a new base
- C. Rebase deletes commits permanently
- D. Merge only works on the main branch

Q36. Which command creates and switches to a new branch in one step?

- A. git branch new-branch
- B. git checkout -b new-branch
- C. git switch old-branch
- D. git merge new-branch

Q37. What does 'git pull' do?

- A. Only fetches remote changes without merging
- B. Fetches and merges remote changes into the current branch
- C. Pushes local commits to remote
- D. Deletes local commits

SQL

Q38. Which query correctly finds the Nth highest salary using LIMIT/OFFSET (MySQL)?

- A. SELECT salary FROM emp ORDER BY salary ASC LIMIT 1
- B. SELECT DISTINCT salary FROM emp ORDER BY salary DESC LIMIT 1 OFFSET (N-1)
- C. SELECT MAX(salary) FROM emp
- D. SELECT salary FROM emp WHERE salary = N

Q39. What is the key difference between TRUNCATE and DELETE?

- A. No difference

- B. TRUNCATE is DDL and resets identity/can't be rolled back easily; DELETE is DML and can use WHERE with rollback
- C. DELETE removes the table structure
- D. TRUNCATE only works on views

Q40. Which JOIN returns all rows from both tables, with NULLs where there's no match?

- A. INNER JOIN
- B. LEFT JOIN
- C. RIGHT JOIN
- D. FULL OUTER JOIN

Q41. What does the GROUP BY clause require when selecting non-aggregated columns?

- A. Nothing extra
- B. Those columns must appear in the GROUP BY clause
- C. An ORDER BY clause
- D. A subquery

Q42. What is a primary key constraint?

- A. Allows duplicate NULLs
- B. Uniquely identifies each row and disallows NULLs
- C. Speeds up only INSERT statements
- D. Is optional in every table

Q43. Which clause filters groups after aggregation, unlike WHERE?

- A. HAVING
- B. WHERE
- C. ORDER BY
- D. GROUP BY

Testing Tools & Web Basics

Q44. In JUnit, which annotation marks a method to run before every test case?

- A. @Test
- B. @BeforeEach
- C. @AfterAll
- D. @Disabled

Q45. What is the primary purpose of Mockito in unit testing?

- A. To run integration tests only
- B. To create mock objects and isolate the unit under test from dependencies
- C. To manage database migrations
- D. To compile Java code

Q46. What HTTP status code indicates a successful resource creation via POST?

- A. 200
- B. 201
- C. 204
- D. 400

Q47. What's the key difference between GET and POST?

- A. GET modifies server data, POST retrieves it

- B. GET retrieves data and is idempotent; POST submits data and can change server state
- C. They are functionally identical
- D. POST cannot carry a request body

Q48. Which HTML tag is used to create a form?

- A. <div>
- B. <form>
- C. <input>
- D. <section>

Q49. What does assertEquals(expected, actual) do in a JUnit test?

- A. Logs a warning if values differ
- B. Fails the test if expected and actual values are not equal
- C. Only works with Strings
- D. Skips the test

ANSWER KEY & EXPLANATIONS

Core Java & OOP

Q1: B. final

final on a method blocks overriding; on a class it blocks inheritance.

Q2: B. 2

Integer division truncates toward zero.

Q3: C. HashSet

Set implementations reject duplicates by contract.

Q4: C. friend

'friend' is a C++ concept, not Java.

Q5: B. It belongs to the class, shared across all instances

Static fields are class-level, not per-object.

Q6: B. Iterable

Iterable provides the iterator() the for-each loop uses.

Q7: B. Two or more threads waiting on locks the other holds

Circular wait on locks is the classic deadlock condition.

Design Patterns

Q8: B. Only one instance of a class exists with global access

Singleton restricts instantiation to exactly one object.

Q9: B. Factory

Factory pattern delegates object creation logic.

Q10: B. Notifying multiple dependents when an object's state changes

Observer implements a publish-subscribe relationship.

Q11: B. Allows incompatible interfaces to work together

Adapter converts one interface into another expected by the client.

Q12: A. Decorator

Decorator wraps objects to add responsibilities at runtime.

Q13: A. An object has many optional constructor parameters

Builder avoids telescoping constructors for complex objects.

Spring Core, Maven & Hibernate

Q14: B. A design pattern where objects receive dependencies rather than creating them

DI is the core IoC mechanism Spring is built around.

Q15: B. pom.xml

pom.xml (Project Object Model) is Maven's core config file.

Q16: B. A single unit of work between the application and the database

A Session wraps a JDBC connection for one unit of work.

Q17: C. @Component

@Component is the generic stereotype; the others are specializations of it.

Q18: B. The Session

First-level cache is Session-scoped and enabled by default.

Q19: C. mvn package

mvn package builds the artifact; install also copies it to the local repo.

Spring Boot & Spring REST

Q20: B. Auto-configuration and embedded servers reduce boilerplate setup

Spring Boot's core value is convention-over-configuration and starters.

Q21: A. @RestController

@RestController returns response bodies directly, no view resolution.

Q22: B. @Value

@Value reads a property placeholder into a field.

Q23: B. POST

PostMapping is a shortcut for @RequestMapping(method = POST).

Q24: B. Automatically injects a matching Spring bean

It resolves and injects a dependency by type (or qualifier).

Q25: B. spring-boot-starter-web

starter-web brings in Spring MVC and an embedded Tomcat.

React

Q26: B. A syntax extension letting you write HTML-like code in JavaScript

JSX compiles to React.createElement calls.

Q27: B. useState

useState returns a state value and setter for functional components.

Q28: B. A change in state or props

React re-renders on state/prop changes via its reconciliation process.

Q29: B. To perform side effects like data fetching after render

useEffect handles side effects tied to the component lifecycle.

Q30: B. Read-only data passed from parent to child components

Props flow one-way, parent to child, and are immutable within the child.

Q31: B. Rendering performance by minimizing direct DOM updates

React diff's the virtual DOM and batches minimal real-DOM updates.

Git

Q32: B. git add

git add moves changes to the staging area.

Q33: B. Copies a remote repository to your local machine

Clone creates a full local copy of a remote repo.

Q34: B. git log

git log lists commits in reverse chronological order.

Q35: B. Merge preserves history with a merge commit; rebase rewrites commit history onto a new base

Merge creates a merge commit; rebase replays commits linearly.

Q36: B. git checkout -b new-branch

-b flag creates the branch then checks it out.

Q37: B. Fetches and merges remote changes into the current branch

git pull = git fetch + git merge in one step.

SQL

Q38: B. SELECT DISTINCT salary FROM emp ORDER BY salary DESC LIMIT 1 OFFSET (N-1)

DESC order with OFFSET (N-1) skips the top N-1 salaries.

Q39: B. TRUNCATE is DDL and resets identity/can't be rolled back easily; DELETE is DML and can use WHERE with rollback

TRUNCATE is a DDL operation, generally faster and non-selective.

Q40: D. FULL OUTER JOIN

FULL OUTER JOIN combines LEFT and RIGHT JOIN results.

Q41: B. Those columns must appear in the GROUP BY clause

SQL requires non-aggregated selected columns to be grouped.

Q42: B. Uniquely identifies each row and disallows NULLs

Primary keys enforce uniqueness and NOT NULL.

Q43: A. HAVING

HAVING filters on aggregated results; WHERE filters rows before grouping.

Testing Tools & Web Basics

Q44: B. @BeforeEach

@BeforeEach runs setup logic before each individual test.

Q45: B. To create mock objects and isolate the unit under test from dependencies

Mockito mocks dependencies so tests focus on one unit's logic.

Q46: B. 201

201 Created is the standard response for successful resource creation.

Q47: B. GET retrieves data and is idempotent; POST submits data and can change server state

GET is safe/idempotent for retrieval; POST typically creates/changes state.

Q48: B. <form>

<form> wraps form controls and defines submission behavior.

Q49: B. Fails the test if expected and actual values are not equal

It's the core assertion that fails a test on mismatch.